

Retail Fashion Store Performance Dashboard – Analysis Report

Overview

The objective of this dashboard is to analyse the performance of a Retail fashion store across sales, customers, products and stores. The dashboard was designed to provide insights into sales performance, customer behaviour, product performance, and store efficiency. Interactive visualizations, calculated columns and measures to identify various key business insights, KPI cards, slicers, drill-through, drill down functionality, and multiple chart types, buttons, bookmarks like different methods were used to support data-driven decision-making.

Data Collection

<https://www.kaggle.com/datasets/vanpatangan/retail-fashion-data>

Data Cleaning

The study also involves data cleaning and preparation, including handling missing values, to ensure accurate and reliable insights using PivotTable analysis

1.Sales_data Sheet

Columns are: transaction_id,date, product_id, store_id, customer_id, quantity, discount, returned

1.In the sales data sheet, column quality, column profile, and column distribution were checked using Power Query.

1844 customer IDs were missing.

Some discount values were also missing.

2.Duplicates were removed from the dataset.

3.The data types of all columns were verified and standardized as follows

4.Column Data Type checked.

transaction_id-String

sales date-Date

product_id-String

store_id-String

customer_id-String

quantity-Integer

discount-Decimal

returned-Integer

5. At first the missing Customer IDs were replaced as UNKNOWN because Customer ID is a mandatory unique identifier and should not be imputed. But on creating the pivot tables,

these records created wrong categorization in the age group analysis, since the rows with missing IDs could not be reliably attributed to valid customers and cannot fetch actual

customer wise data from the other tables. So these records were excluded from the final dataset to ensure accuracy.

6. Column header format is also standardized by renaming the headers.

7. Missing discount values were handled using the imputation method. The average discount value was calculated using statistical analysis and used to replace the missing values.

8. Text formatting was also standardized using the "Capitalize Each Word" option.

New Columns Added

1. Year

Extracted year from transaction date.

2. Month_Name

Extracted month name for monthly sales analysis.

3. Return_Status - Added conditional column.

Converted:

1 → Returned

0 → Not Returned

4. Sales_Amount - Custom column by applying formula

$\text{quantity} \times \text{list_price}$

Represents total sales before discount.

5. Total_Cost - Custom column by applying formula

$\text{quantity} \times \text{cost_price}$

Represents total product cost.

9. The columns list_price and cost_price were not originally present in the sales data sheet.

These columns were imported from the product_data sheet using the Merge Queries option in Power Query.

During data integration, some transaction records contained invalid product IDs that were not present in the product master table.

These unmatched records generated null values after merging.

So such rows were identified and removed by filtering null values from merged columns,

2. product_data Sheet

This sheet contains product-related information of the fashion store.

Original Columns: product_id, category, color, size, season, supplier, cost_price, list_price

1. In the sales data sheet, column quality, column profile, and column distribution were checked using Power Query.

2. Column headers were standardized and duplicates were removed.

3. In the color column, missing values were filled using the mode imputation method.

To identify the mode:

The "Group By" operation was used.

Color frequencies were counted.

The grouped values were sorted in descending order.

The most frequent color identified was "Green".

The missing values were then replaced with "Green" using the Replace Values option.

4. Invalid category values such as "???" were identified and corrected using mode imputation.

The frequency distribution of the category column was analyzed using:

Group By

Count operations in Power Query

The category with the highest frequency (mode) was identified and used to replace the invalid category entries.

5. Data Type Standardization

6.cost_price was converted to Decimal Number.

list_price was converted to Decimal Number

New Columns Added

1.Profit_Percentage

Used to identify highly profitable products.

2.Price_Category

based on list_price

This helps analyze customer purchasing patterns.

3. customer_data Sheet

This sheet contains customer demographic details.

Original Columns

customer_id

age

gender

city

email

1.Column quality, column profile, and column distribution were checked.

2.Duplicates were removed and column headers were standardized.

3. Missing gender values were replaced using the imputation method.

New Columns Added

1. Age_Group

Customers grouped into:

18–25

26–35

36–45

46–60

60+

Used for customer segmentation and purchasing behavior analysis by using conditional formatting.

4. store_data Sheet

This sheet contains information about store branches.

Original Columns:store_id, store_name, region, store_size_m2

This was a small sized data.so column quality,distribution and profile were not checked.

column headers were standardized.

New Columns Added

1. Store_Size_Category

if store_size_m2 <300 then Category is small, 300–700 then Medium and if store_size_m2>700 then Large

Dashboard Pages and Analysis

DAX Measures created in Sales Data

Total Revenue

Total Manufacturing Cost

Total Profit

Profit Margin%

Average discount

Count of customers, products, orders , returned orders

New column just to show the discount amount and net profit are also added.

DAX Measures created in Customer Data

Top customer, Top purchased gender, top purchased age group, top purchased city were calculated.

DAX Measures created in Product Data

Top purchased product, Return rate, top category and the highest sales season are calculated.

DAX Measures created in Store Data

Top performing region, store, and also top store category are calculated.

Also revenue per metre is also calculated to know how efficiently a store is using its floor space to generate revenue.

Overview

This page contains the

- Total Sales
- Total Profit
- Manufacturing Cost
- Number of Orders
- Number of Customers
- Number of Products
- Returned Orders

The timeline helps to identify the growth of business over the years by analysing the total sales amount, manufacturing cost, profit gained, number of orders, customers who trusted the store over the years.

Sales Analysis

A year selection slicer is added in the left side to check the

1.The line chart helps to analyse the monthly sales trend by giving the transaction date in the x axis and the sum of sales amount in the y axis and also the year in the legend.

Drill-Down Functionality

Users can drill down from:

Year → Quarter → Month as the date hierarchy is given in the x axis.

This helps detailed sales analysis.

2.Category-wise sales comparison is analysed using a clustered bar chart by considering the sales amount and product category.

3.Regional wise revenue analysed using a tree map by taking region and the sum of sales amount.

4.Profit trend is analysed using area chart by taking the total profit in the y axis and transaction date month in the x axis.

Key Insights

The 2023 marked the highest sales among the years, whereas the 2021 recorded the lowest over the years. Also the Quarter1 and quarter 4 are the lowest performed between the four quarters. Revenue fluctuates across months, indicating seasonal demand.

Accessories generated the highest revenue.

Algarve and Lisbon contribute significantly more revenue than others, whereas the Porto contributed the least.

May and July showed highest profit than other months.

Customer Analysis

KPI Cards helped to identify the top purchased customer(C011508), gender, city and age group.

1.This donut chart analyses revenue contribution across customer age groups.

2.This pie chart compares total sales across gender categories
Male -34.8% , Other - 33.0% , Female - 32.2%

3.Stacked column chart helps to compares sales performance across cities.

Lisbon, Faro, Braga, Coimbra, Porto are the cities participated in the sales.

1. The age group and gender matrix provides a detailed breakdown of sales by both Age Group and Gender. Wise sales amount contribution.

Key Insights

1.The 46–60 age group contributes approximately **27% of total sales**, making the most valuable contribution.

2. Sales distribution is relatively balanced among genders. Although Male customers contribute slightly more revenue compared to others.

3. Sales are evenly distributed across cities like Lisbon, Faro, Braga, Coimbra, Porto. But Lisbon leading overall in the revenue compared to others.

4. Age group 46–60 generates the highest total sales. Male customers within the 46–60 segment contribute the highest revenue. The 60+ age group contributes the lowest sales among all segments.

Product Analysis

KPI Cards Analysis showed total 30,630 Products Sold and P022324 as Top Purchased Product and Accessories as Top Purchased Category and Winter as Top purchased Season.

1.Profit by Category stacked column chart compares total profit generated by each product category. Accessories -1.40M, Dresses-1.36M, Shoes-1.34M, Tops-1.32M, Bottoms-1.32M

2. Returns by Category area chart evaluates product return behaviour across categories

Accessories-10,019

Dresses-9,473

Shoes-9,470

Bottoms-9,460

Tops-9,347.

3. Seasonal Performance

This funnel chart compares sales performance across different seasons.

4. Supplier Contribution: This pie chart evaluates the contribution of different suppliers to overall revenue.

Supplier D-25.13%

Supplier B-25.07%

Supplier A-24.96%

Supplier C-24.83%

Key Insights

1. Accessories generate the highest profit. All categories contribute relatively similar profit levels.

2. Even though accessories generate the highest profit, Accessories experience the highest number of returns also. Tops have the lowest return count.

3. The top purchased season is winter, still the summer season generated highest profit compared to other seasons. However, business keeps almost similar performance on all seasons.

4. Supplier contributions are almost evenly distributed and no single supplier dominates the sales.

4. Store Analysis

KPI Cards Analysis gave the Top Performing Region as Algarve and store as Faro Outlet and store efficiency tells how effectively store floor space generates revenue. Top Performed Store Type is small among the large, medium and small.

1. Stacked bar chart shows the profit by region

2. The scatter chart analyzes the relationship between store size and profitability.

3. This line chart compares sales performance across store categories small, medium and large.

4. Profit by Store clustered column chart compares profitability at individual store level.

Key Insights

1. Algarve generates the highest profit. Lisbon closely follows Algarve. Profitability is distributed fairly evenly across all regions.
2. Larger stores do not necessarily generate significantly higher profits. Smaller stores performed well than larger stores.
3. Small stores generate the highest sales. Large stores perform similarly to small stores. Medium-sized stores underperformed.
4. Faro Outlet is the most profitable store. Lisbon Flagship performs almost equally well. Store performance is relatively balanced.

Drill Through Functionality

A dedicated **Transaction Details** page was created so that users can directly view the category wise, city wise, supplier wise and store wise transactions from the Sales Analysis, Customer analysis, product analysis and Store analysis and navigate directly to the detailed transaction records related to the selected ones using the drill through property.

The transaction details page displays: Transaction ID ,Customer ID ,Product ID, Category, Region, Sales Amount , Net Profit, Return Status

Bookmark Functionality: Separate bookmarks were created for:

Accessories Transaction Details

Shoes Transaction Details

Dresses Transaction Details

Bottoms Transaction Details

Top Transaction Details

Top Region Transaction Details

